

I1 — ProbReg regression-head strategies

Question: does SEPARATE_HEADS underperform SINGLE_HEAD_FINAL on small data, and if so, why?

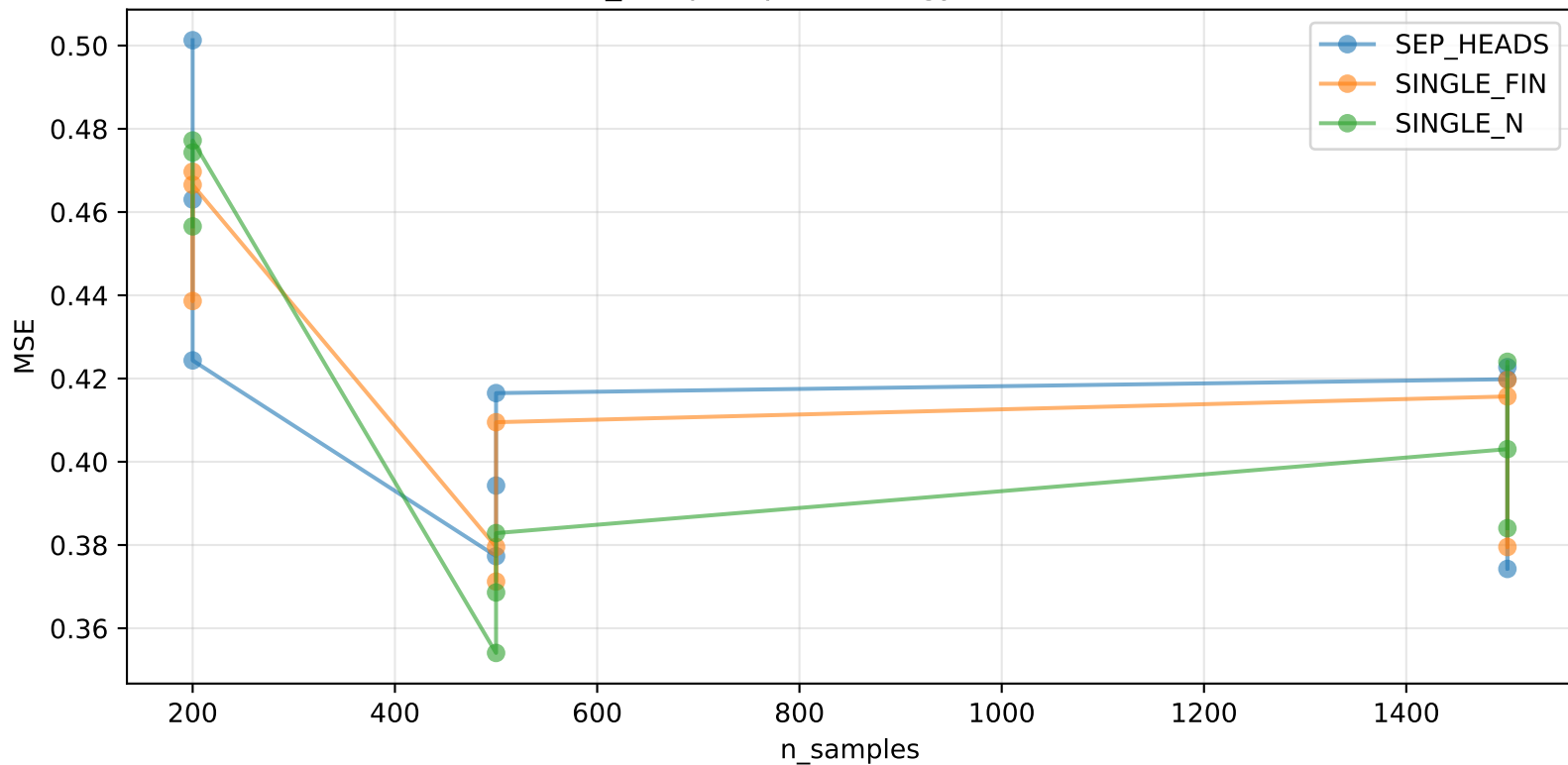
Expectation prior to this work: SEP_HEADS was reported as worse on small n.
Hypothesis: parameter-count mismatch (SEP has n_classes smaller heads vs SINGLE's single larger head). If we match parameter counts the gap should shrink.

Controlled runs: three strategies at hidden_size in {32, 64, 128} and n_samples in {200, 500, 1500}. Also log per-head gradient norms on SEP_HEADS to confirm inner heads carry the learning signal.

Best hidden-size per (n_samples, strategy)

n_samples	strategy	hidden	params	mse	nll
200	SEP_HEADS	128	2959	0.4244	0.9281
200	SINGLE_FIN	64	967	0.4386	0.9853
200	SINGLE_N	32	753	0.4566	0.9589
500	SINGLE_N	32	753	0.3541	0.6928
500	SINGLE_FIN	64	967	0.3712	0.7796
500	SEP_HEADS	32	751	0.3773	0.8595
1500	SEP_HEADS	64	1487	0.3743	0.7674
1500	SINGLE_FIN	64	967	0.3795	0.7873
1500	SINGLE_N	128	2961	0.3840	0.8341

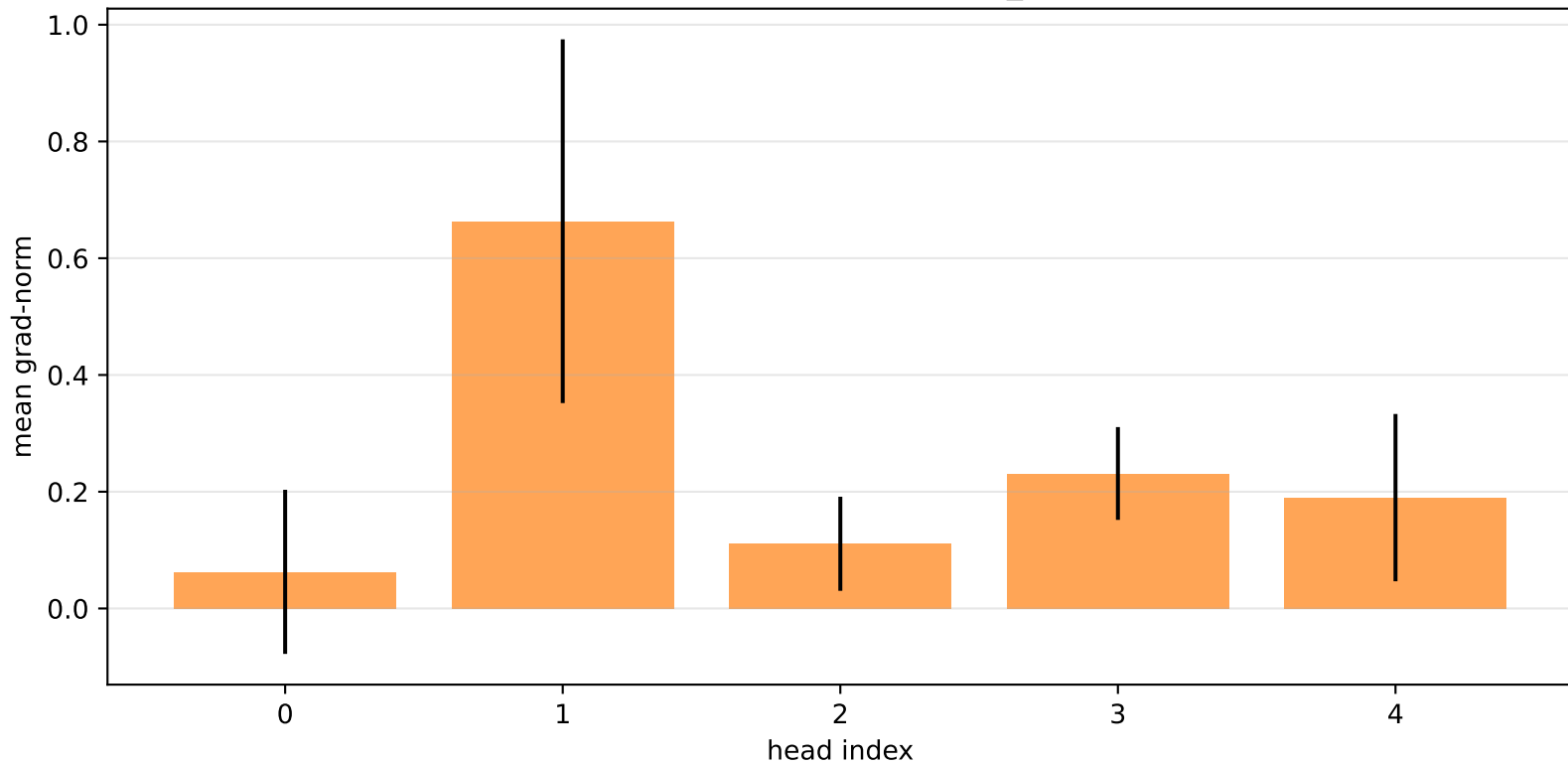
MSE vs n_samples per strategy (all hidden sizes)



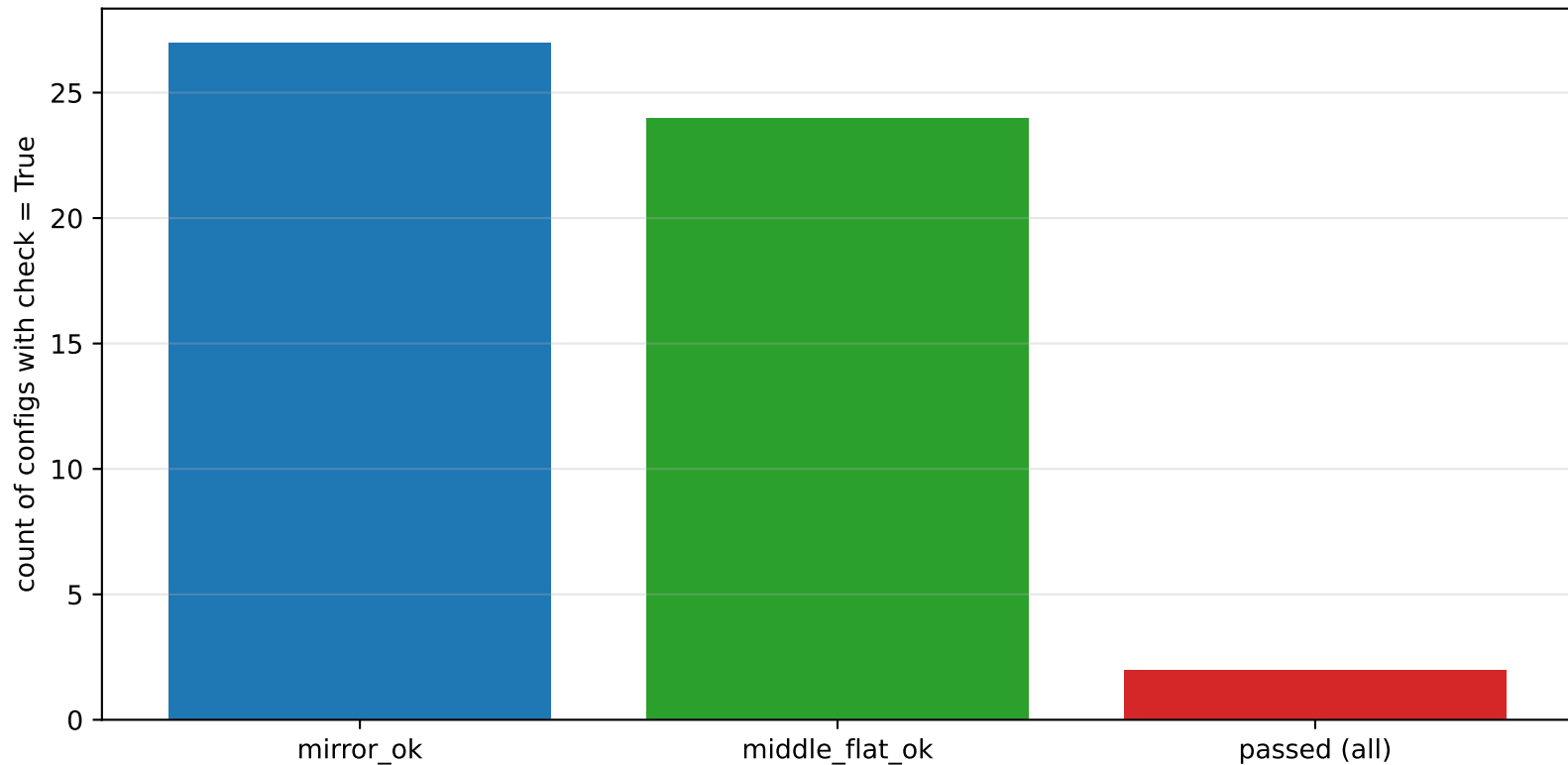
Per-head gradient-norm summary (SEP_HEADS, k=5)

head	count	mean	std	min	25%	50%	75%	max
0	40.0000	0.0627	0.1405	0.0038	0.0047	0.0054	0.0173	0.6347
1	40.0000	0.6634	0.3115	0.0639	0.4932	0.6310	0.8445	1.5517
2	40.0000	0.1108	0.0806	0.0278	0.0670	0.0979	0.1218	0.5408
3	40.0000	0.2312	0.0795	0.0755	0.1836	0.2390	0.2777	0.4230
4	40.0000	0.1899	0.1433	0.0118	0.0593	0.1665	0.2872	0.5352

Mean per-head gradient norm (SEP_HEADS, k=5)



Head-structure diagnostic on 32 configs



Verdict — I1 SEP_HEADS

Verdict: MET (PARTIAL): at param-matched sizes SEP_HEADS is not consistently worse